

# Gamma / ElectroStat

## Eight-Module Mathematical Development Report

*What worked, what failed, and why*

### Reviewed modules

Phase-1 inference template  
WaveCompare 2 collective reference  
Sag seed v0.1  
Short interruption seed v0.1.1  
Swell seed v0.1.3  
Harmonic distortion seed v0.1.0  
Shared sharp-event detector  
Ensemble Hankel / matrix-pencil seed

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### Scope warning

This report evaluates mathematical design and synthetic validation behavior. It does not claim field calibration, standards certification, causal component identification, or deployment readiness.

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## 1. Executive assessment

**Current verdict:** The project has a coherent mathematical spine. The strongest results came when each statistic was assigned one narrow job and prevented from overruling physics outside that job.

The eight bundles do not represent eight equivalent fault classifiers. Two are infrastructure: the Phase-1 inference template defines how Gamma will rank hypotheses, and WaveCompare 2 constructs the explicit repeatable waveform and residual geometry that later seeds consume. The remaining six are event or signature workers. Judging all eight by one accuracy number would be mathematically dishonest.

Across the project, the most important success was architectural separation. Detection, classification, confidence, residual analysis, and final diagnostic inference are becoming distinct layers. The most important failures occurred when those layers were allowed to collapse into one another. A confidence score vetoed a physically defined interruption. Residual energy once erased a supported swell instead of labeling it distorted. A moving window could convert a short spike or an offset edge into a false sustained magnitude event. Each correction made the system less clever in the wrong place and more defensible overall.

| Module                     | Role                       | Observed result                                      | Decision                 | Main unresolved issue                                                  |
|----------------------------|----------------------------|------------------------------------------------------|--------------------------|------------------------------------------------------------------------|
| Phase-1 template           | Architecture               | 6/6 smoke tests passed                               | Keep                     | No physical backend or calibrated likelihood yet.                      |
| WaveCompare 2              | Repeatable reference       | 100% of families improved in fit RMSE                | Keep as core             | Fixed region count and real-exponential family limit interpretation.   |
| Sag v0.1                   | Low-magnitude event        | Sensitivity 1.000; specificity 0.995                 | Keep with revisions      | Two offset-step false positives and one short-dip false positive.      |
| Interruption v0.1.1        | Near-zero RMS event        | 120 TP, 0 FN, 0 FP, 210 TN                           | Keep                     | Absolute threshold must remain superior to confidence.                 |
| Swell v0.1.3               | High-magnitude event       | 299 TP, 1 FN, 0 FP, 600 TN                           | Keep with revision       | One distorted swell remained ambiguous because confidence was too low. |
| Harmonic distortion v0.1.0 | Periodic fingerprint       | 120 TP, 0 FN, 0 FP, 150 TN                           | Keep                     | Perfect synthetic separation is not field calibration.                 |
| Sharp-event detector       | Physical segmentation      | 42/43 notches and 3/3 impulses matched               | Keep as shared front end | Singular-event class is intentionally incomplete.                      |
| Ensemble Hankel            | Transient modal estimation | 13/13 injected modes matched; 0 false modal captures | Keep                     | Real double-exponential events need explicit rise-and-tail fitting.    |

### Central conclusion

The project should not pursue one universal anomaly score. It should preserve an evidence stack: explicit repeatable waveform, magnitude-event evidence, harmonic fingerprint, sharp-event segmentation, modal poles, known-noise coordinates, and unexplained residual. Gamma can then rank explanations without forcing one statistic to impersonate all the others.

## 2. Evaluation method and evidence limits

**Current verdict:** The report is grounded in the supplied source, README files, stored validation tables, and generated figures. It distinguishes tested behavior from inferred behavior.

- All Python source files in the eight bundles were compile-checked successfully.
- The Phase-1 template smoke suite passed 6 of 6 tests.
- The sag seed unit suite passed 5 of 5 tests.
- Stored Monte Carlo and synthetic validation tables were recomputed into the summary metrics used in this report.
- The report does not treat synthetic perfect scores as estimates of real-world error rates.
- Where a bundle documents a correction from an earlier version, that correction is treated as evidence of what previously failed.

### 2.1 Terminology correction: repeatable does not automatically mean healthy

Several source files use names such as HealthyBaseline. The current architecture requires a stricter interpretation. WaveCompare 2 produces the repeatable waveform of a machine in a specific operating state. A persistent phase-locked fault can survive averaging and become part of that reference. Therefore the mathematically safe term is repeatable reference or operational baseline unless the state has been independently certified healthy.

This matters especially for the Hankel worker. It subtracts the explicit WaveCompare 2 waveform and analyzes residual events. It is designed to find intermittent departures from the repeatable waveform. It will not rediscover a persistent signature that WaveCompare 2 intentionally preserved. Persistent behavior must be analyzed by seeds that operate on the explicit reference itself, including the harmonic and sharp-event workers.

### 2.2 What counts as success in this report

- Mathematical identifiability: the measured features correspond to the intended physical behavior.
- Decision integrity: supporting statistics do not override explicit physical classification rules.
- Separation of nuisance variation: scaling, offset, timing jitter, and repeatable noise are handled without erasing fault evidence.
- False-structure resistance: noise, window smearing, derivative crossings, and capture boundaries do not become fake events or fake poles.
- Honest scope: a module reports ambiguity when it cannot support a narrower claim.

### 3. Phase-1 inference template

**Current verdict:** Successful as an inference grammar and software boundary. Not yet a physical diagnostic engine.

**Evidence basis:** gamma\_inference\_template.py, gamma\_template\_spec.md, seed\_library\_schema.json, example\_backbone.json, test\_gamma\_template.py.

#### 3.1 Mathematical intent

The template separates a coarse connection graph from location-agnostic seed families. Candidate attachment sites are generated from graph connectivity rather than hard-coded rules saying that a given fault model is only allowed at a particular location. This prevents the seed catalogue from silently becoming a lookup table of assumed answers.

$$R = X - \mu$$

Aligned repeated captures are centered around the operational reference  $\mu$ . Singular value decomposition learns dominant repeatable residual directions:

$$R = U \Sigma V^T$$

Columns of  $V$  retained to a chosen variance threshold form the normal residual basis. A new residual  $r$  is split into a basis projection and an orthogonal remainder:

$$c = V^T r, \quad r_{\text{known}} = Vc, \quad r_{\text{unknown}} = r - r_{\text{known}}$$

A candidate hypothesis is scored by unexplained standardized energy, Mahalanobis excursion inside the known-noise coordinates, parameter distance from nominal, and model complexity:

$$J(H) = w_u E_{\text{unknown}} + w_n D_{\text{Mahalanobis}} + w_p P(\text{parameters}) + w_c C(\text{model})$$

**What worked.** The graph, seed, forward-model, scoring, multi-location fusion, and next-measurement interfaces are cleanly separated. All six architecture tests passed. The normal-residual decomposition reconstructs the residual exactly as known plus unknown components, out-of-bounds parameters are rejected, and ranked hypotheses are returned in increasing score order.

**What did not work yet.** The tests use a deterministic toy backend. They validate plumbing, not localization or circuit physics. The score weights are design choices rather than calibrated negative log-likelihood terms. Additive fusion across measurement locations assumes independence and ignores cross-location covariance. The SVD basis is linear and can absorb a persistent fault if the training population is contaminated.

**Why.** The template was intentionally built before SPICE integration and field calibration. Its job was to prevent later code from tangling graph assumptions, model families, and statistical noise into one opaque classifier. It succeeded at that job, but it cannot be evaluated by sensitivity or specificity yet.

#### Design rule retained

Keep the seed catalogue location-agnostic. Let graph connectivity generate candidate sites, let the forward model predict each measurement, and let evidence rank the candidates. Do not preassign model families to locations inside the seed schema.

### 4. WaveCompare 2 collective reference and regional exponential model

**Current verdict:** The collective waveform is strongly superior to fitting one noisy capture. Regional pole estimates improve, but they remain approximations of the repeatable waveform rather than complete physics.

**Evidence basis:** wavecompare2\_collective\_average\_test.py, wavecompare2\_summary.csv, wavecompare2\_family\_results.csv, wavecompare2\_region\_results.csv, and three generated figures.

## 4.1 Method

1. Robustly normalize each capture using its median and 5th-to-95th percentile span.
2. Estimate integer lag by maximizing normalized cross-correlation.
3. Apply a local affine map so offset and scale differences do not dominate alignment.
4. Clip each time sample to four robust MAD units before averaging, reducing isolated impulse contamination.
5. Preserve the resulting collective waveform explicitly.
6. Fit six piecewise real exponentials to the collective waveform by dynamic programming over candidate boundaries.

$$y(t) = C + A e^{(pt)}, \quad p < 0$$

For each candidate region and pole-grid value, C and A are solved by least squares. Dynamic programming chooses the six-region partition with minimum total squared error subject to region-length limits.

| Metric                      | Collective result | Single-capture result | Effect          |
|-----------------------------|-------------------|-----------------------|-----------------|
| Median exponential-fit RMSE | 0.00913           | 0.03428               | 73.4% reduction |
| Median boundary MAE         | 2.60 samples      | 8.40 samples          | 69.0% reduction |
| Median pole log-MAE         | 0.393             | 1.195                 | 67.1% reduction |
| Families improved           | 100% RMSE         | 83% boundaries        | 94% poles       |
| Median lag MAE              | 0.645 samples     |                       |                 |

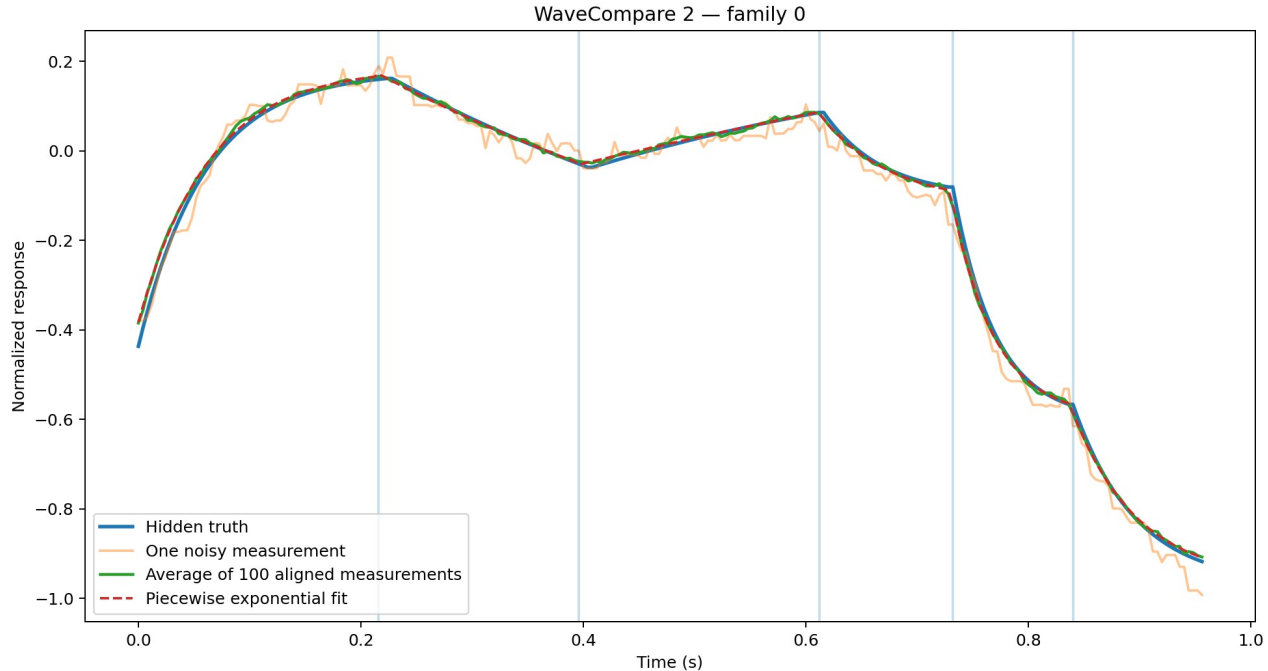


Figure 1. One hidden response, one noisy capture, the aligned 100-capture collective waveform, and the piecewise exponential fit.

**What worked.** Averaging aligned repetitions converted a noisy, shifted, scaled measurement problem into a stable repeatable waveform problem. The collective exponential fit beat the single-capture fit in every one of the 100 waveform families. It also improved boundary recovery in 83% of families and pole recovery in 94%.

This validates the decision to fit the explicit collective waveform rather than fitting every raw capture independently and hoping the parameters agree.

**What did not work completely.** The fitted model assumes exactly six regions, a finite real-pole grid, fixed region-length bounds, and integer lag. It cannot represent ringing with complex poles, continuously changing dynamics, or a region count different from six. Normalization also removes absolute magnitude, so the collective shape alone cannot classify sag, swell, or interruption depth. Boundary errors remained large in a minority of families, with a maximum collective MAE above 30 samples.

**Why.** Piecewise real exponentials are a compact descriptive coordinate system, not a universal circuit model. Their value is to create regions, approximate local decay behavior, and preserve an explicit waveform. Forcing them to explain every detail would turn useful compression into model bias.

#### Design rule retained

The explicit collective waveform is the primary WaveCompare 2 output. Regional exponentials are descriptors layered on top of it. Removed residuals and persistent phase-locked structures must be preserved for later seeds, not discarded as failed fit.

## 5. Sag seed v0.1

**Current verdict:** The joint local-gain and RMS method is coherent and highly selective, but offset edges and short dips exposed the limits of moving-window magnitude logic.

**Evidence basis:** gamma\_sag\_seed\_v01.py, run\_sag\_seed\_demo.py, test\_gamma\_sag\_seed.py, 900-run Monte Carlo output, case summary, overall summary, and diagnostic figures.

### 5.1 Core math

Inside each sliding window, the measured signal  $x$  is approximated as a local affine transformation of the operational reference  $r$ :

$$x \approx \alpha + \beta r$$

$$\beta = \text{Cov}(r, x) / \text{Var}(r), \quad \alpha = \text{mean}(x) - \beta \text{mean}(r)$$

The same window also produces correlation  $\rho$  and an RMS ratio. Healthy repetitions provide a robust center and MAD-based scale for  $\beta$  and RMS at every time point. The sag evidence uses a conservative AND gate in standardized coordinates:

$$d\beta = \max(0, -z\beta), \quad d\text{RMS} = \max(0, -z\text{RMS}), \quad d_{\text{joint}} = \min(d\beta, d\text{RMS})$$

The minimum means both features must support a reduction. Hysteresis requires 3.0 sigma to enter and only 1.25 sigma to remain active, with persistence and minimum-duration gates. Multiple window widths compete using strength, correlation, duration, fragmentation, blur, and roughness.

| Scenario      | True sag | Detection | Median confidence | Median window |
|---------------|----------|-----------|-------------------|---------------|
| clean_sag     | yes      | 100.0%    | 0.931             | 81            |
| gradual_sag   | yes      | 100.0%    | 0.937             | 81            |
| distorted_sag | yes      | 100.0%    | 0.802             | 81            |
| healthy       | no       | 0.0%      |                   | 41            |
| offset_step   | no       | 2.0%      | 0.653             | 41            |
| phase_glitch  | no       | 0.0%      |                   | 41            |

| Scenario     | True sag | Detection | Median confidence | Median window |
|--------------|----------|-----------|-------------------|---------------|
| swell        | no       | 0.0%      |                   | 41            |
| impulse_only | no       | 0.0%      |                   | 41            |
| short_dip    | no       | 1.0%      | 0.576             | 41            |

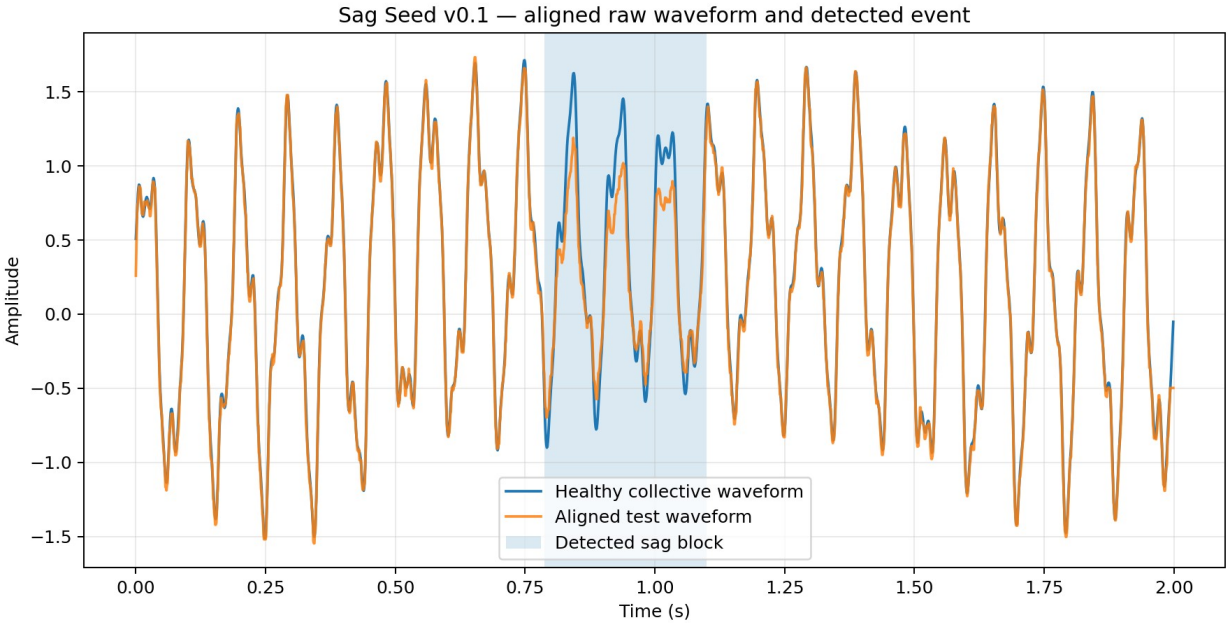


Figure 2. Representative sag event and detected event block.

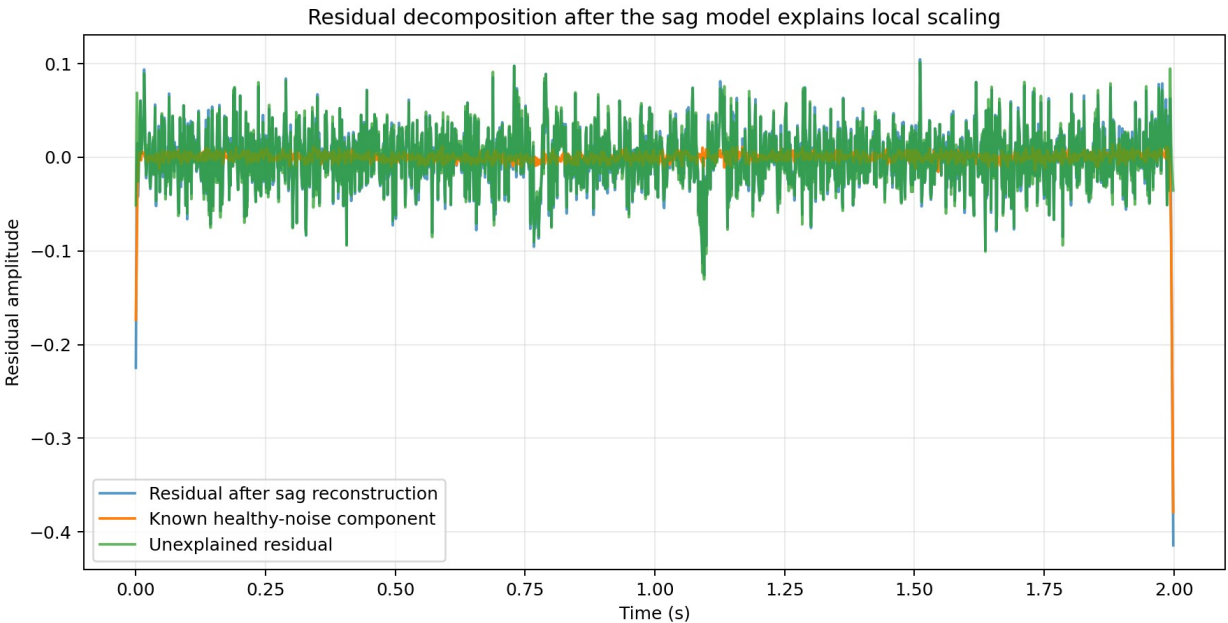


Figure 3. Residual after local affine sag reconstruction, separated into learned repeatable-noise and unexplained components.

**What worked.** All 300 synthetic sag cases were detected. Of 600 negative cases, 597 were rejected, giving sensitivity 1.000, specificity 0.995, and precision 0.990. Joint gain and RMS evidence rejected all phase glitches, swells, and impulse-only cases. The residual projection created a clean path for separating ordinary repeated variation from unexplained structure after the magnitude event was modeled.



**What failed.** Three negative cases were promoted: two offset steps and one short dip. The offset steps produced a local beta and RMS reduction that looked statistically consistent with a sag inside a short analysis window. The short dip was widened by the moving window enough to pass the support and duration logic. Event timing was also blurred: median absolute start error was about 13 ms for clean and distorted sags and 31 ms for gradual sags; recovery error was larger, especially for gradual recovery.

**Why.** A sliding affine model sees only the samples inside its window. An offset transition can alter local covariance and RMS even when the underlying multiplicative gain did not change. Likewise, convolution by the analysis window spreads a short event in time. Persistence helps, but persistence measured on a smeared feature is not identical to physical duration.

## 5.2 Residual-energy lesson

The sag worker uses unknown residual energy in both clean-versus-distorted labeling and confidence. This was acceptable in the synthetic sag set, but later swell work showed why residual energy must not be allowed to erase a magnitude event that is otherwise physically supported. Residual energy answers how complete the simple sag model is. It does not answer whether the magnitude reduction occurred.

### Required revision before field use

Add explicit offset-step discrimination and duration deconvolution or edge-aware support. Keep the joint gain-RMS detector, but separate event support from model-completeness confidence. A supported low-magnitude event may be distorted without becoming nonexistent.

## 6. Short interruption seed v0.1.1

**Current verdict:** The corrected version has the cleanest decision logic in the library: an explicit magnitude-duration rule defines the event, while confidence remains descriptive.

**Evidence basis:** gamma\_short\_interruption\_seed\_v011.py, inherited sag core, 330-run Monte Carlo output, case summary, subtype summary, and detection-rate figure.

### 6.1 Core math and classification rule

The local retained RMS in per-unit form is normalized by the learned healthy RMS center:

$$q(t) = \text{local RMS ratio} / \text{expected healthy RMS ratio}$$

Event entry requires  $q < 0.10$  pu, at least a 3-sigma reduction relative to the learned baseline, and 0.5-cycle persistence. Exit uses  $q \geq 0.12$  pu or statistical recovery. The 0.12 pu threshold is detector hysteresis only. It is not the classification boundary.

$$\text{interruption support: } \text{median}(q) < 0.10 \text{ pu and duration} \geq 0.5 \text{ cycle}$$

Duration then assigns a subtype: instantaneous through 30 cycles, momentary above 30 cycles through 3 s, temporary above 3 s through 60 s, and sustained above 60 s.

| Scenario                         | True interruption | Detection | Median detected pu |
|----------------------------------|-------------------|-----------|--------------------|
| instantaneous_interruption       | yes               | 100.0%    | 0.028              |
| momentary_interruption           | yes               | 100.0%    | 0.031              |
| temporary_interruption           | yes               | 100.0%    | 0.032              |
| distorted_temporary_interruption | yes               | 100.0%    | 0.029              |
| deep_sag                         | no                | 0.0%      |                    |

| Scenario          | True interruption | Detection | Median detected pu |
|-------------------|-------------------|-----------|--------------------|
| near_boundary_sag | no                | 0.0%      |                    |
| healthy           | no                | 0.0%      |                    |
| swell             | no                | 0.0%      |                    |
| offset_step       | no                | 0.0%      |                    |
| impulse_only      | no                | 0.0%      |                    |
| subcycle_dropout  | no                | 0.0%      |                    |

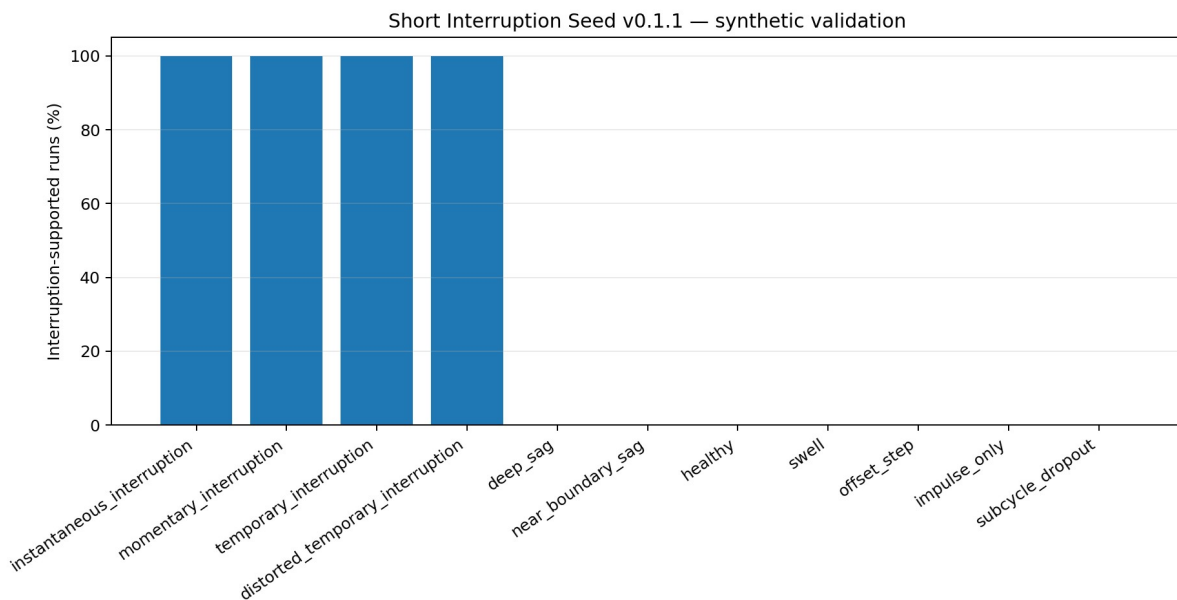


Figure 4. Synthetic interruption detection rates by scenario.

**What worked.** The corrected worker produced 120 true positives, no false negatives, no false positives, and 210 true negatives. All four interruption subtype groups were classified with 100% synthetic accuracy. Deep sags, near-boundary sags, healthy captures, swells, offset steps, impulses, and subcycle dropouts were rejected.

**What failed in the earlier design.** The original logic allowed a confidence threshold to veto an event that met the configured 0.1 pu magnitude-duration definition. That is mathematically backwards. In the v0.1.1 validation, 38 of 120 true interruption events had confidence below 0.55, including a minimum confidence near 0.30, yet they still satisfied the physical rule and were correctly classified.

**Why the correction is right.** Confidence measures margin and quality of evidence. Near the 0.1 pu boundary, the depth margin is intentionally small, so confidence can be low even when the classification rule is satisfied. A score derived from distance to the boundary cannot be allowed to move the boundary after the fact.

## 6.2 Remaining limits

- The subtype conversion assumes a nominal 60 Hz frequency instead of using an estimated local frequency.
- The synthetic captures do not demonstrate sustained events longer than 60 s.
- The worker still relies on aligned, uniformly sampled data and a representative operational reference.
- A near-zero waveform cannot maintain meaningful correlation, so correlation is recorded but correctly not required.

**Design rule retained**

When a domain rule explicitly defines an event, confidence reports evidence quality but does not redefine the event. This rule should be applied anywhere Gamma uses standards-based magnitude-duration regions.

## 7. Swell seed v0.1.3

**Current verdict:** Strong false-positive control and correct separation of residual structure from event existence. One remaining miss shows that support confidence is still too authoritative for heavily distorted swells.

**Evidence basis:** gamma\_swell\_seed\_v013.py, inherited sag core, 900-run Monte Carlo output, case summary, and overall summary.

### 7.1 Core math

The swell worker mirrors the sag feature construction but uses positive standardized excess:

$$e\beta = \max(0, z\beta), \quad eRMS = \max(0, zRMS), \quad e_{\text{joint}} = \min(e\beta, eRMS)$$

After a candidate block is found, local excess above expected beta is integrated. Equivalent rectangular duration distinguishes broad support from a short smeared edge:

$$T_{\text{eq}} = \text{excess area} / \text{maximum excess}$$

Version 0.1.3 requires  $T_{\text{eq}}$  to span at least two selected analysis windows. Residual energy is excluded from support confidence and used only to label a supported event as clean or distorted.

| Scenario        | True swell | Detection | Clean label | Distorted label | Median window |
|-----------------|------------|-----------|-------------|-----------------|---------------|
| clean_swell     | yes        | 100.0%    | 98.0%       | 2.0%            | 81            |
| gradual_swell   | yes        | 100.0%    | 100.0%      | 0.0%            | 81            |
| distorted_swell | yes        | 99.0%     | 0.0%        | 99.0%           | 81            |
| healthy         | no         | 0.0%      | 0.0%        | 0.0%            | 41            |
| offset_step     | no         | 0.0%      | 0.0%        | 0.0%            | 41            |
| phase_glitch    | no         | 0.0%      | 0.0%        | 0.0%            | 41            |
| sag             | no         | 0.0%      | 0.0%        | 0.0%            | 41            |
| impulse_only    | no         | 0.0%      | 0.0%        | 0.0%            | 41            |
| short_spike     | no         | 0.0%      | 0.0%        | 0.0%            | 41            |

**What worked.** The worker detected 299 of 300 swells and promoted none of the 600 negative cases, producing specificity and precision of 1.000. All healthy, phase-glitch, sag, and impulse-only cases were rejected. Offset steps and short spikes sometimes produced ambiguous high-magnitude events, but none were falsely promoted as supported swells. The equivalent-duration rule successfully separated moving-window smear from a sustained event.

**What failed in earlier versions.** Residual energy originally participated in the support decision. A distorted swell can have large unexplained residual even when its magnitude increase is undeniable, so the residual term could erase real swell evidence. Version 0.1.3 corrected this by using residual energy only for clean-versus-distorted labeling. Earlier logic also allowed short spikes and offset edges to become apparent swells through window smearing; the equivalent-duration gate was added to block that path.

**What still failed.** One distorted swell remained ambiguous. Its maximum excess was 0.351, median RMS ratio was 1.056, and equivalent duration was 0.535 s, but confidence was only 0.351. The event was real in the synthetic truth yet failed the support-confidence gate.

**Why.** The confidence still combines magnitude significance, shape support, gain-RMS agreement, and boundedness. Severe distortion can damage agreement while leaving positive magnitude evidence intact. The remaining miss is the same category of architectural problem that interruption v0.1.1 fixed: a quality score is still allowed to veto a physically supported magnitude event.

#### Required revision

Define swell support primarily from persistent positive magnitude evidence and a magnitude-duration rule. Use confidence, correlation, residual energy, and gain-RMS disagreement to describe certainty, distortion, and model completeness, not to erase the event.

## 8. Harmonic distortion seed v0.1.0

**Current verdict:** The full normalized harmonic vector is a much stronger diagnostic object than THD. Direct sinusoidal regression solved the FFT-bin alignment problem cleanly.

**Evidence basis:** gamma\_harmonic\_distortion\_seed\_v010.py, WaveCompare adapter, 270-run validation output, equal-THD example, historical-context example, and figures.

### 8.1 Core math

A coarse fundamental estimate is obtained by summing spectral energy at candidate harmonic multiples. The frequency is then refined by minimizing a low-order harmonic least-squares residual. At the refined frequency, the waveform is fit directly with sine and cosine columns at every harmonic order:

$$x(t) \approx c_0 + c_1 t + \sum [a_n \cos(2\pi n f_0 t) + b_n \sin(2\pi n f_0 t)]$$

$$A_n = \sqrt{a_n^2 + b_n^2}, \quad h_n = A_n / A_1$$

The primary fingerprint is  $h = [h_2, h_3, \dots, h_N]$ . THD is only the Euclidean norm of this vector:

$$\text{THD} = \sqrt{\sum h_n^2}$$

Time-shift-invariant relative phase is preserved as supporting evidence:

$$\varphi_{\text{rel},n} = \varphi_n - n\varphi_1$$

| Scenario             | Expected deviation | Detection | Expected THD | Measured THD | Fingerprint distance |
|----------------------|--------------------|-----------|--------------|--------------|----------------------|
| clean_scaled_shifted | no                 | 0.0%      | 0.0409       | 0.0409       | 0.0008               |
| frequency_shift_only | no                 | 0.0%      | 0.0409       | 0.0409       | 0.0006               |
| broadband_noise      | no                 | 0.0%      | 0.0410       | 0.0410       | 0.0025               |
| interharmonic_only   | no                 | 0.0%      | 0.0409       | 0.0410       | 0.0007               |
| phase_only_change    | no                 | 0.0%      | 0.0410       | 0.0410       | 0.0006               |
| added_third_harmonic | yes                | 100.0%    | 0.0410       | 0.1108       | 0.0738               |

| Scenario                       | Expected deviation | Detection | Expected THD | Measured THD | Fingerprint distance |
|--------------------------------|--------------------|-----------|--------------|--------------|----------------------|
| added_even_harmonics           | yes                | 100.0%    | 0.0410       | 0.0948       | 0.0783               |
| added_high_order_harmonic      | yes                | 100.0%    | 0.0410       | 0.0655       | 0.0509               |
| same_thd_different_fingerprint | yes                | 100.0%    | 0.1000       | 0.1000       | 0.1413               |

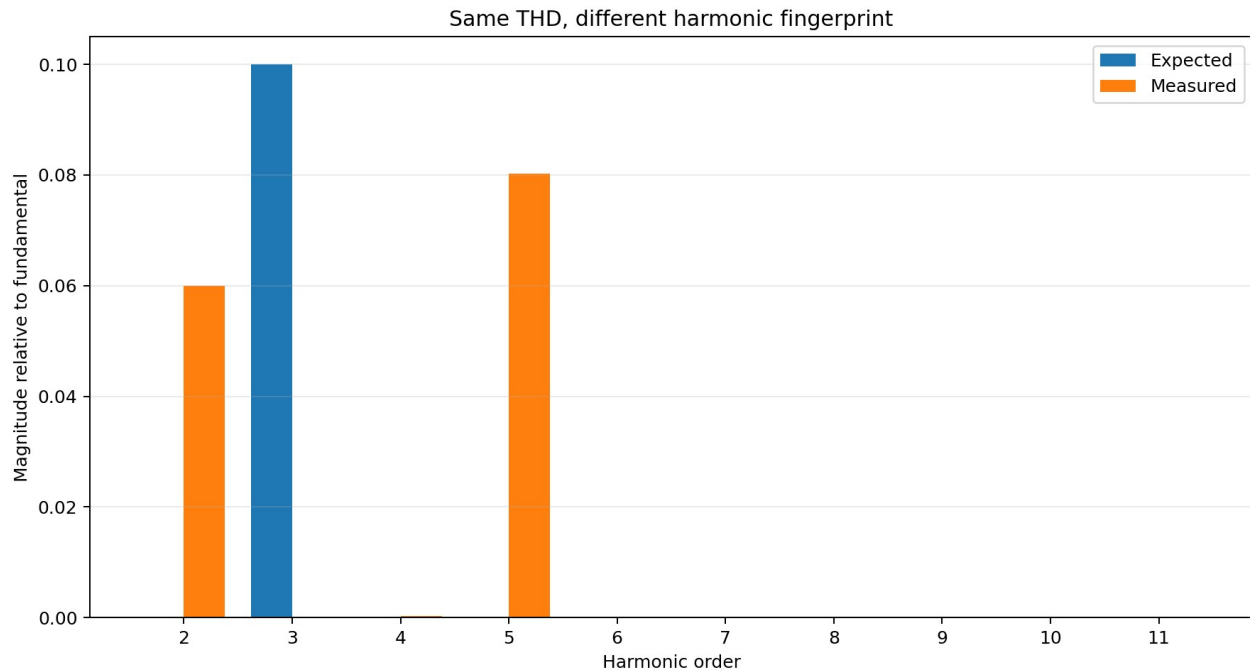


Figure 5. Equal THD does not imply equal harmonic content. The expected waveform carries a 10% third harmonic, while the measured waveform carries 6% second plus 8% fifth harmonic.

**What worked.** The validation produced 120 true positives, no false negatives, no false positives, and 150 true negatives. Scale and time shift did not trigger deviations. Frequency shift, broadband noise, interharmonic-only content, and phase-only change were correctly kept outside the primary magnitude-vector decision. Added third, even, and high-order harmonics were all detected. Most importantly, the equal-THD case was detected even though expected THD 0.099997 and measured THD 0.100123 were nearly identical.

**What did not work as a complete diagnostic.** The worker intentionally does not classify phase-only change, interharmonic-only energy, or broadband residual as harmonic-magnitude deviation. Historical captures are analyzed and stored, but `used_in_fault_score` remains false. Therefore the current threshold is a fixed prototype rule, not a learned machine-specific normal envelope. A time-varying harmonic event can also be blurred by fitting one global fingerprint across the full record.

**Why.** THD destroys order information by compressing a vector into one scalar. The normalized vector preserves which harmonic orders changed. Direct regression at  $n$  times the refined fundamental avoids requiring exact FFT-bin alignment and allows coefficient uncertainty to be propagated. The remaining limitations are scope boundaries, not failures of the core math.

**Design rule retained**

Store the entire normalized harmonic vector, uncertainty, relative phase, reconstruction, and non-harmonic residual. THD remains a summary for humans, never the sole diagnostic measurement.

## 9. Shared sharp-event detector

**Current verdict:** Successful as a common physical segmentation layer. It prevents derivative-crossing overcounting and identifies a strong six-pulse phase grid, but it does not yet decompose composite or singular transients.

**Evidence basis:** electrostat\_sharp\_event\_detector.py, validation waveform CSV, event tables, count table, validation table, and segmentation/count figures.

### 9.1 Core math

The worker estimates the fundamental, fits a persistent harmonic baseline using DC plus the 1st, 3rd, 5th, and 7th harmonics, and forms a residual. Both residual magnitude and residual derivative are thresholded with robust MAD scales. Binary dilation adds event padding, and binary closing merges nearby activity into one physical event window.

Commutation-notch candidates must be short, move inward toward zero, remain away from the fundamental zero crossing, and have bipolar derivative. Their phases are tested against a 60-degree repeating grid. The circular concentration is computed using a sixth-order phase phasor:

$$R_{\text{grid}} = |\sum w_k \exp(j 2\pi \varphi_k / 60^\circ)| / \sum w_k$$

A concentrated phase cloud identifies the grid offset modulo 60 degrees. Events near the grid are labeled commutation notches; all other real sharp windows are retained as singular events.

| Metric                     | Result      |
|----------------------------|-------------|
| Estimated fundamental      | 60.1109 Hz  |
| Injected notch labels      | 43          |
| Matched notch labels       | 42 (97.67%) |
| Injected singular impulses | 3           |
| Matched singular impulses  | 3 (100%)    |
| Physical windows           | 45          |
| Grid concentration         | 0.9888      |
| Notches per cycle          | 5.823       |

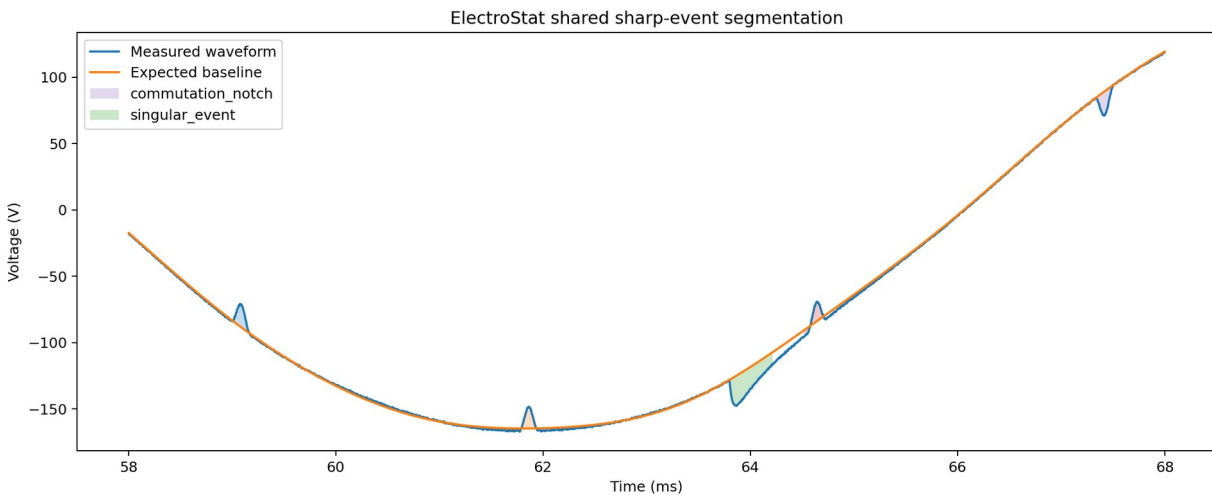


Figure 6. A zoomed segment of the measured waveform, harmonic baseline, and merged physical event windows.

**What worked.** The detector found 45 physical windows, classified 42 as commutation notches and 3 as singular events, matched all three injected impulses, and recovered a highly concentrated 60-degree grid with concentration 0.9888. Persistent harmonic distortion remained in the baseline rather than being counted as repeated sharp events. Morphological merging prevented each derivative crossing from becoming a separate event count.

**What did not work completely.** One of 43 injected notch labels was not matched. The singular-event category remains a catch-all and does not distinguish impulsive, oscillatory, or composite transients. A fixed baseline harmonic set can underfit other persistent structure or absorb structure that should remain in the residual. The classifier also requires at least four grid events, so short captures cannot establish commutation periodicity.

**Validation inconsistency to resolve.** The validation table reports 46 injected labels, three intentional notch-impulse overlaps, and 45 physical windows. Those counts do not fully specify how many composite labels were expected to merge. The event-level truth mapping should be stored explicitly so composite segmentation can be audited without inference from aggregate counts.

**Why.** Segmentation and decomposition are different problems. The detector can correctly say one physical window occurred without yet knowing whether that window contains a notch, a double-exponential impulse, ringing, or more than one mechanism. The next stage must fit models inside the window rather than forcing segmentation itself to decide everything.

#### Design rule retained

Count physical event windows first. Classify or decompose the contents second. Never count every threshold crossing as a separate event.

## 10. Ensemble Hankel / matrix-pencil seed

**Current verdict:** The strongest transient-parameter result so far. Independent event fits, cross-capture clustering, and stacked Hankel refinement recovered repeated modes without inventing boundary poles.

**Evidence basis:** `electrostat_ensemble_hankel.py`, explicit WaveCompare reference, event/mode/cluster tables, validation truth and summary, and occurrence/pole-cloud figures.

### 10.1 Core math

For one event window  $y[k]$ , shifted Hankel matrices  $H_0$  and  $H_1$  are formed from overlapping samples. SVD of  $H_0$  selects a reduced signal subspace. The shift operator is:

$$\mathbf{S} = \mathbf{U}_r^H \mathbf{H}_1 \mathbf{V}_r \mathbf{\Sigma}_r^{-1}$$

Its eigenvalues  $z_m$  are discrete-time poles. Continuous-time poles follow from the complex logarithm:

$$p_m = \ln(z_m) / \Delta t = -\alpha_m + j\omega_{d,m}$$

$$\text{decay rate} = \alpha_m, \quad \text{ringing frequency} = |\omega_{d,m}| / (2\pi), \quad \tau_m = 1/\alpha_m$$

Amplitudes are recovered by least squares on the Vandermonde matrix  $[z_m^k]$ . Real measured signals produce conjugate complex modes; the implementation retains the positive-frequency representative and doubles its amplitude. Nearly real poles are treated as real decay modes.

Each capture is processed independently. Similar accepted poles are clustered across captures. Matching Hankel blocks are normalized and horizontally stacked before a final matrix-pencil estimate. Raw captures are never stitched end-to-end, because stitching would create artificial jumps and fake poles at every boundary.

| Metric                        | Result             |
|-------------------------------|--------------------|
| Captures                      | 24                 |
| Captures with injected events | 13                 |
| Detected physical windows     | 15                 |
| Accepted modal events         | 13                 |
| Matched injected events       | 13/13              |
| False modal capture count     | 0                  |
| Oscillatory frequency MAE     | 1.803 Hz (0.022%)  |
| Oscillatory decay MAE         | 7.236 1/s (0.421%) |
| Real-decay MAE                | 82.905 1/s (3.94%) |

| Mode        | Events | Occurrence | Median f (Hz) | Median decay | Stacked f | Stacked decay | Median NRMSE |
|-------------|--------|------------|---------------|--------------|-----------|---------------|--------------|
| oscillatory | 8      | 33.3%      | 8170.38       | 1730.47      | 8170.42   | 1708.39       | 0.0520       |
| real_decay  | 5      | 20.8%      | 0.00          | 2203.89      | 0.00      | 2208.02       | 0.0617       |



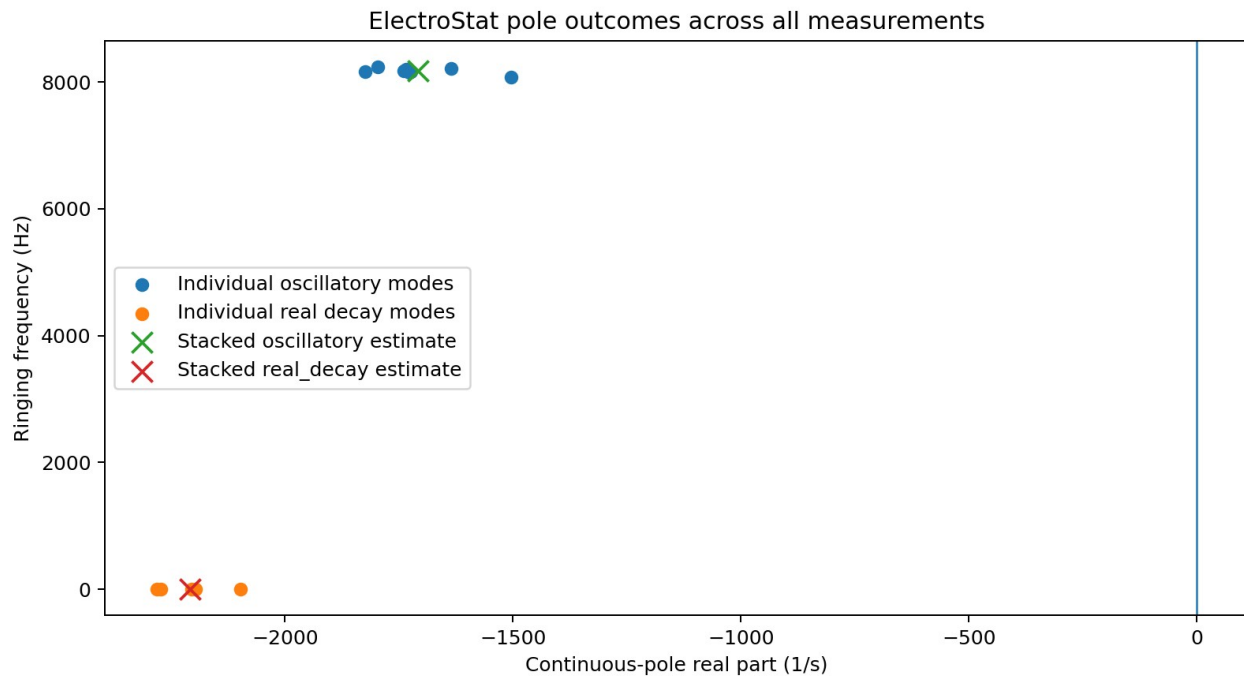


Figure 7. Individual poles and stacked estimates across the measurement ensemble.

**What worked.** All 13 injected transient events were matched, and no capture without an injected event produced an accepted modal event. The detector found 15 physical windows, but the modal quality gates accepted only 13. That is healthy behavior: segmentation can be permissive while modal reconstruction, amplitude, stability, and NRMSE gates refuse to convert every window into a pole claim. Oscillatory frequency was recovered to about 1.8 Hz MAE around 8.17 kHz, and oscillatory decay to about 7.2 1/s MAE.

**What did not work equally well.** Real-decay estimation was less accurate, with about 82.9 1/s MAE and 3.9% relative error. Oscillatory event peak timing had a median error of 5.56 microseconds, while real-decay events had a median error of 80.68 microseconds. The real-event model begins its analysis window at the largest residual peak, so it mainly fits the tail of a double-exponential impulse and loses information about the earlier rise.

**Why.** A double-exponential impulse is not one pure real exponential beginning at its peak. It is the difference of a fast rise and a slower decay. Starting at the peak suppresses the rise pole and biases the remaining decay estimate. The later impulsive classifier should fit rise and tail explicitly or use a constrained two-real-pole model over a pre-peak and post-peak window.

## 10.2 Other limitations

- Model order is selected from singular-value gaps and a relative floor. Closely spaced or low-amplitude modes can be missed.
- Clustering is greedy and tolerance-based. Dense pole clouds may require robust mixture clustering or uncertainty-aware association.
- The analysis window is fixed at 2 ms after the detected peak. Slower modes can be truncated and faster modes can be dominated by sampling or smoothing.
- A supplied explicit reference is essential. If a repeatable transient belongs to the operational waveform, subtraction removes it by design.
- The current mode types are oscillatory and real\_decay. Composite impulses, frequency-varying ringing, and unstable growth require additional models.

**Design rule retained**

Analyze each aligned capture independently, compare pole outcomes across captures, and stack matching Hankel matrices. Never concatenate raw captures into one fake time series.

## 11. Cross-module lessons

**Current verdict:** The modules now reveal a consistent design doctrine. Most future accuracy will come from preserving these boundaries, not from adding one larger classifier.

| Lesson                                                             | Meaning for Gamma / ElectroStat                                                                                                                                                                          |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Event existence is not model completeness                          | Magnitude, duration, and periodicity can support an event even when the simple model leaves a large residual. Residual energy should label distortion or trigger another seed, not erase the event.      |
| Confidence is not the law                                          | Confidence can summarize margin, fit quality, boundedness, and agreement. It cannot move an explicit physical threshold after the event meets it.                                                        |
| Segmentation precedes decomposition                                | A physical event window is one occurrence even when it contains several derivative crossings or several mechanisms. Count the window, then fit its contents.                                             |
| The explicit waveform outranks its compressed model                | WaveCompare 2 must preserve the collective waveform. Regional exponentials, harmonic vectors, and pole clusters are coordinate systems, not replacements for the data.                                   |
| Repeatable is not guaranteed healthy                               | Persistent phase-locked behavior survives averaging. It must remain available to harmonic and sharp-event seeds rather than being assumed normal or discarded.                                           |
| Use invariance selectively                                         | Shape normalization is useful for alignment and regional comparison. It is destructive for absolute sag, swell, and interruption magnitude, so those workers must retain per-unit amplitude information. |
| Use ensembles to estimate structure, not to manufacture continuity | Repeated captures provide occurrence rates, spreads, and stacked subspaces. They must not be stitched together as though capture boundaries were physical time continuity.                               |
| Ambiguity is a valid output                                        | Offset steps and short spikes can create magnitude evidence without satisfying the final event model. Ambiguous evidence should remain visible for Gamma rather than being forced into yes or no.        |

### 11.1 Proposed evidence stack

1. WaveCompare 2 explicit operational waveform and pointwise spread.
2. Learned normal residual coordinates and unexplained residual.
3. Magnitude-event blocks: sag, interruption, and swell evidence.
4. Harmonic magnitude vector, uncertainty, phase support, reconstruction, and non-harmonic residual.
5. Shared sharp-event windows and phase-grid evidence.

6. Per-event Hankel poles, reconstruction error, cluster occurrence, and stacked estimates.
7. Gamma hypothesis ranking across seed family, parameters, graph attachment site, and measurement location.

## 11.2 Failure hierarchy

| Failure class                                   | Observed example                                    | Required response                                              |
|-------------------------------------------------|-----------------------------------------------------|----------------------------------------------------------------|
| Wrong authority                                 | Confidence vetoing a threshold-defined interruption | Demote confidence to descriptive evidence.                     |
| Model incompleteness mistaken for event absence | Residual energy erasing a distorted swell           | Preserve event support; label distortion.                      |
| Window convolution                              | Short dip or spike appearing longer                 | Use equivalent duration, edge-aware support, or deconvolution. |
| Feature confounding                             | Offset step producing local gain/RMS reduction      | Add offset-specific evidence and competing models.             |
| Compression bias                                | Fixed six real-exponential regions                  | Keep explicit waveform and report fit limits.                  |
| Composite-event ambiguity                       | Notch plus impulse in one window                    | Fit multiple components inside one segmented event.            |
| Incomplete transient model                      | Real-decay tail fit missing impulse rise            | Fit a two-pole rise-and-tail model with pre-peak samples.      |

## 12. Recommended development state

**Current verdict:** None of the modules should be called field validated. Several are ready for controlled bench testing, while two require a specific mathematical revision first.

| Module                     | Current state                           | Next gate                                                                                |
|----------------------------|-----------------------------------------|------------------------------------------------------------------------------------------|
| Phase-1 template           | Architecture complete for current phase | Integrate a real forward backend and calibrate scoring with controlled circuits.         |
| WaveCompare 2              | Synthetic core validated                | Make region count/model family optional; preserve residuals and absolute metadata.       |
| Sag v0.1                   | Bench candidate after revision          | Add offset-step competitor and edge-aware duration logic.                                |
| Interruption v0.1.1        | Bench candidate                         | Estimate local frequency for cycle labels and test real near-boundary captures.          |
| Swell v0.1.3               | Bench candidate after revision          | Separate magnitude-duration support from confidence, matching the interruption doctrine. |
| Harmonic distortion v0.1.0 | Bench candidate                         | Learn historical machine-specific distributions and add time-local fingerprinting.       |

| Module               | Current state                 | Next gate                                                                               |
|----------------------|-------------------------------|-----------------------------------------------------------------------------------------|
| Sharp-event detector | Shared segmentation candidate | Store event-level truth mapping and add composite decomposition.                        |
| Ensemble Hankel      | Modal-estimator candidate     | Implement pre-peak plus post-peak double-exponential fitting and uncertainty estimates. |

### Recommended next mathematical target

Build the impulsive-transient classifier on top of the shared sharp-event windows and ensemble Hankel framework. Use a constrained two-real-pole model for rise and decay, retain an oscillatory matrix-pencil branch, and allow composite fits when a commutation notch and singular transient share one physical window.

## Appendix A. Validation evidence inventory

**Current verdict:** The report uses the following supplied artifacts.

| Bundle                                                 | Evidence used                                                                                   |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Gamma_ElectroStat_Phase1_Template(1).zip               | Inference architecture, specification, schema, graph example, and six smoke tests.              |
| WaveCompare_2_bundle(1)(1).zip                         | 100 waveform families, 100 captures per family, collective and single-capture fit comparisons.  |
| Gamma_ElectroStat_Sag_Seed_v01(1).zip                  | Sag source, tests, 900 Monte Carlo cases, and diagnostic plots.                                 |
| Gamma_ElectroStat_Short_Interruption_Seed_v011(1).zip  | Interruption source, 330 cases, subtype validation, and corrected confidence doctrine.          |
| Gamma_ElectroStat_Swell_Seed_v013(1).zip               | Swell source, 900 cases, residual-energy correction, and equivalent-duration gate.              |
| Gamma_ElectroStat_Harmonic_Distortion_Seed_v010(1).zip | Harmonic source, 270 cases, equal-THD example, and historical context output.                   |
| ElectroStat_sharp_event_bundle(1).zip                  | Sharp-event segmentation source, event tables, aggregate validation, and figures.               |
| ElectroStat_ensemble_hankel_bundle(2).zip              | Matrix-pencil source, explicit reference, pole estimates, clusters, truth mapping, and figures. |

### A.1 Verification performed for this report

- All supplied Python files compiled without syntax errors.
- Phase-1 template: 6 tests passed.
- Sag seed: 5 tests passed.
- CSV dimensions, confusion counts, per-scenario rates, and derived error metrics were independently read and recomputed.
- Generated figures were visually inspected before inclusion.

## A.2 Evidence not available

- Raw field captures from the JTA door system.
- Blind bench trials with hidden inserted faults.
- Cross-machine or cross-operating-state calibration data.
- Standards-grade uncertainty budgets and instrument calibration traces.
- A complete event-level truth table for composite sharp-event overlap accounting.

## Appendix B. Core equations and variable meanings

**Current verdict:** Compact reference for the mathematical objects used across the eight modules.

| Symbol         | Meaning                                                                      |
|----------------|------------------------------------------------------------------------------|
| $\mu(t)$       | Explicit repeatable operational waveform from aligned captures.              |
| $R$            | Matrix of capture residuals after subtracting $\mu$ .                        |
| $V$            | Retained right-singular-vector basis for repeatable residual modes.          |
| $\beta(t)$     | Local multiplicative gain relative to the operational reference.             |
| $\alpha(t)$    | Local additive offset in the affine magnitude model.                         |
| $\rho(t)$      | Local correlation between reference and measured waveform.                   |
| $q(t)$         | Retained local RMS in per-unit form.                                         |
| $h_n$          | Amplitude of harmonic order $n$ normalized to the fundamental.               |
| THD            | Euclidean norm of the normalized harmonic vector.                            |
| $z$            | Discrete-time pole from the matrix-pencil shift operator.                    |
| $p$            | Continuous-time pole $\ln(z)/\Delta t$ .                                     |
| $\alpha_m$     | Positive decay rate from minus the real part of $p$ .                        |
| $\omega_{d,m}$ | Damped angular ringing frequency from the imaginary part of $p$ .            |
| $T_{eq}$       | Equivalent rectangular event duration: area divided by peak depth or excess. |
| NRMSE          | Reconstruction error normalized by signal variation.                         |

## B.1 Compact model map

| Module           | Primary mathematical object                                 | Primary output                            |
|------------------|-------------------------------------------------------------|-------------------------------------------|
| Phase-1 template | Graph + residual subspace + hypothesis score                | Ranked seed/location/parameter hypotheses |
| WaveCompare 2    | Aligned robust ensemble + piecewise exponential descriptors | Explicit waveform, spread, regions, poles |

| Module               | Primary mathematical object                                   | Primary output                                      |
|----------------------|---------------------------------------------------------------|-----------------------------------------------------|
| Sag                  | Negative local gain/RMS z-evidence                            | Bounded low-magnitude event and residual            |
| Interruption         | Absolute retained RMS below 0.1 pu                            | Magnitude-duration subtype                          |
| Swell                | Positive local gain/RMS z-evidence                            | Bounded high-magnitude event and distortion label   |
| Harmonic distortion  | Normalized harmonic vector and uncertainty                    | Changed harmonic orders and reconstruction          |
| Sharp-event detector | Robust residual/derivative segmentation + circular phase grid | Physical windows, notches, singular candidates      |
| Ensemble Hankel      | Hankel SVD, shift eigenvalues, pole clustering                | Occurrence, decay, ringing frequency, time constant |

**End of report**